

Correction note

Prepended 21 September 2026 to the record as originally rendered, which follows unchanged.

Correction note — 21 September 2026 (D-42). The needs-review figure in this document, 168 of 430 (39.1%), was measured while a classifier defect was caching failed API batches as abstentions. How many of the 168 were affected cannot be recovered, because the cache does not timestamp its entries, so read the figure as an upper bound. At 442 rows the corrected figure is **128 (29.0%)**. The document below is left as written. See [docs/d42_failure_record.md](#) for the cause, the evidence and the fix.

Network Visualiser — Status and Pipeline, Day 15

Current board, runtime pipeline, and the Phase 2 QA checkpoint definition of done

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1. Purpose and currency

This document records where the project stands on **Day 15** of the re-baselined 19-21 focused-day plan, and defines what “done” means for the current session.

It supersedes Sections 3 and 9 of `project_state_decision_log.pdf`. That document remains authoritative on *decisions and reasoning* — the Phase 1.5 pivot, the classification ceiling, the approaches ruled out with evidence, the privacy posture. It is **stale on status**: it records the board as of 17 August, when P2.5 was the next task and 2.5 of 9 items were complete. Six items have shipped since. Do not read the board from it.

Last committed working session: **28 August 2026 (Day 14)**. Nothing has been committed since. Verify with `git --no-pager log --oneline -10` before trusting this document from that date onward.

2. Position in the plan

Plan	19-21 focused days to public launch (re-baselined from the original 17)
Today	Day 15
Phase	Phase 2 — MVP dashboard
Phase 2 build work	Complete. All build items shipped and committed.
Remaining in Phase 2	QA checkpoint (today), P2.9a, P2.9
Standing instruction from Day 14	Run the QA checkpoint against every Phase 2 DoD before starting P2.9a and P2.9

3. The pipeline as it stands today

The system is now **two pipelines**, not one. That split is recent, is a consequence of P2.8b, and matters directly to P2.9 — the runtime half is the part that must survive an arbitrary user upload.

3.1 Build-time pipeline (offline batch)

Run once against a `Connections.csv`; produces the canonical node table.

#	Stage	Module	Input	Output
1	Ingestion	linkedin.py, ingestion.py	Connections.csv	Person objects
2	Classification	claude_classifier.py, onet.py	Job titles	SOC major group or abstain
3	Embedding	embeddings.py	profile_text	384-dim vectors
4	Projection	projection.py, projection_3d.py	Vectors	UMAP 2D in-table; 3D sidecar
5	Graph	graph.py	Vectors	Cosine-similarity graph, $\tau = 0.7710$
6	Centrality	centrality.py	Graph	Canonical node table

Canonical handoff: `data/graph/network_nodes.csv` — 430 rows carrying identity, SOC group, UMAP coordinates, centrality metrics, `classifier_tier` provenance, and the `is_uncertain` flag. This file is the contract between the engine and the interface. It is permanently gitignored following the `git-filter-repo` history purge.

3.2 Runtime pipeline (every Streamlit rerun)

`src/dashboard/loader.py` is the **single assembly path** for the display table. There is no second route, and that is enforced rather than asserted.

```
base CSV
└─> 3D sidecar merge      (raises SidecarMismatch with counts;
    |                     never renders a silent subset)
└─> corrections overlay   (state.py – flat dict[int, str] by person_index)
└─> notes overlay        (notes.py – session-scoped)
└─> display table         (derives display_state:
                           classified / needs_review / not_occupation)
```

Enforcement: `tools/check_single_loader.py` parses source with **AST**, not `grep` — adopted after a docstring mention produced a false positive. Four callers were migrated onto the loader: `app.py`, `check_chat_grounding.py`, `check_uncertainty_honesty.py`, `check_summary_safety.py`.

3.3 Dashboard layer

Every module below reads the loader’s display table. None of them re-reads the base CSV.

Module	Responsibility
canvas.py	Pure <code>build_figure(df, match_indices, highlight_index)</code> ; <code>Scatter3d</code> only; fixed 23-colour SOC palette drawn from the taxonomy, never from the data; scene annotations for the selected-node label
search.py	Pure <code>filter_nodes(df, query)</code> across name, role, SOC group name, notes
people_list.py	One component, two modes: search results and the review queue. Skips are session-only and record no verdict
card.py	Profile card; provenance from <code>classifier_tier</code> ; confirmation control (23 groups + “not an occupation”)
state.py	Correction overlay; the only write path for an assignment
notes.py	Per-person free text; <code>on_change</code> callback; empty notes delete their key
enrich.py / self_enrichment.py	Self-enrichment via “About” text; two calls (title-only, title-plus>About) shown side by side; no company slot; no cache
summaries.py	Lazy group summaries on selection; cache keyed on sorted <code>person_index</code> hash; groups <5 show verbatim titles; no names or company sent outbound
chat.py / assistant.py	Two-layer grounded chat: pure core (table in, client injected)

	plus Streamlit glue; prefixed string handles ("p87")
session_io.py	JSON save/restore keyed on identity, not person_index

3.4 Key numbers

Measure	Value
People in the network	430
Distinct job titles	~371
Classified	262 (60.9%)
Flagged "Needs review"	168 (39.1%)
SOC major groups in taxonomy / palette	23
SOC groups actually present in the data	18
Graph threshold τ	0.7710
Modularity Q	0.5224
Components / isolates	13 / 4
UMAP trustworthiness (k=5)	0.9840

The 168 is derived live from the table at every surface. It is hardcoded nowhere. Confirming that remains a checkpoint item.

4. Phase 2 board

4.1 Shipped and committed

Item	Delivered
P2.1	Correction overlay logic (state.py, 9 tests); UI half landed inside P2.5
P2.2	Self-enrichment via "About" text; self_enriched tier
P2.2b	Per-person notes, session-scoped and searchable
P2.3	3D render, visually verified
P2.4 / P2.4b	Search panel and match-set highlight with faded non-matches
P2.5	Profile cards with the P2.1 confirmation control
P2.6a	Group multi-select with dimming; coloured centroid labels
P2.7	Lazy, cached cluster summaries
P2.8	Grounded chat panel
P2.8b	Shared display-table loader with AST enforcement
P2.8c	Session save and restore
P2.8d	Review queue, live result cards, three-column layout

4.2 Deferred out of Phase 2 by amendment

Item	Moved to	Reason
P2.6 SOC-tree zoom	P3.9	Blocked behind detailed-code resolution
2D / 3D canvas toggle	Phase 4	Cheap, but a preference layer on a working canvas
Non-contiguous group centroid labels	Phase 4	Known limitation: labels sit at the lobe mean
Taxonomy skills on profile cards	P3.6 (group-level only)	Amendment 2 — never attributed to an individual

4.3 Remaining in Phase 2

Item	Status	Estimate

Phase 2 QA checkpoint	Today	0.5 day
P2.9a — synthetic 430-row demo table	Not started	0.5 day
P2.9 — Community Cloud deploy, README, zero-crash gate	Not started	1-1.5 days

4.4 Health at last commit

Check	Result
Full test suite	223 passing
<code>check_uncertainty_honesty.py</code>	9 / 9
<code>check_summary_safety.py</code>	11 / 11
<code>check_chat_grounding.py</code>	26 / 26
<code>check_single_loader.py</code>	6 / 6
Git	Clean, pushed

These are **last session's** numbers. Re-running them on current HEAD is checkpoint item 2, not a formality.

5. Definition of done — Day 15 (Phase 2 QA checkpoint)

Today is a checkpoint, not build work. It is done when the following are true.

1. **Every Phase 2 DoD carries an explicit written verdict.** All items in `phase2_4_definitions_of_done.pdf`, plus the three recorded amendments, including the amended criteria: list-driven selection replacing click selection, the enrichment-sourced skills line, and the P2.6 split. Verdicts are recorded against the running app, not from memory or from last session's transcript.
2. **All four harnesses and the full suite re-run green on current HEAD.** Terminal output from Day 14 does not count.
3. **One manual pass through the core interactions, no crash:** load → search → select → correct from the card → correct from the queue → self-enrich → add a note → select a group → read a summary → ask the chat two questions → save → refresh → restore.
4. **The needs-review counter is verified live-derived** at every surface it appears: header, queue, card, coverage panel. Any hardcoded 168 is a defect.
5. **The not-an-occupation count is reconciled.** Two prior sessions report this differently. Establish the live figure from the table and record it.
6. **Failures are logged as named defects with a phase assignment, not fixed inline.** Fixing during a checkpoint is how a checkpoint stops being one.

5.1 Deliberately excluded from today's gate

Two clauses of the written Phase 2 QA checkpoint cannot be tested before a deployment exists, and move to P2.9's own gate:

- “the deployed app renders a real network with grey nodes present and correctable”
- “no real personal data in the public deploy” — this is P2.9a's entire purpose

6. What follows the checkpoint

6.1 P2.9a — synthetic demo table

Generate a synthetic 430-row node table with roughly a 39% unclassifiable cohort, matching the real network's grey-node proportion, and expanded name pools to avoid collision across 430 draws. Scheduled deliberately for immediately before deployment, not during development, so that development ran against real data throughout.

6.2 P2.9 — deployment

Streamlit Community Cloud; boots cleanly from the repo with no real personal data present; full render without crashing; README recording local setup with a real export; O*NET attribution in the README and the dashboard footer; a short

demo recording.

6.3 Phase 3 — team composition and network insight (8-10 focused days)

P3.1 team composition queries; P3.2 complementarity ranking; P3.3 serendipity suggestions via graph traversal; P3.4 top-connector chart from betweenness; P3.5 cluster skill matrices; P3.6 group-level occupational profiles; P3.7 complementarity by occupational distance; P3.9 SOC-tree zoom behind detailed-code resolution. P3.8 (query expansion over notes) is ruled out on clustering, compliance, method, and density grounds.

6.4 Phase 4 — polish and public launch (4-5 days)

Team synergy narratives, UI animations, dark mode, PDF export, onboarding tour, 2D/3D toggle, centroid label fix, performance documentation, public GitHub launch.

7. Rules carried forward

These were learned through failures and are not open for re-litigation.

Classification. The ceiling is the input, not the method. Two-word titles top out at ~60-64% classifiability, measured three convergent ways. Config E is locked: Claude, title-only, no deterministic tier in front, explicit abstention over fabrication. Do not tune further.

Honesty. A blank or flagged node is honest; a confident wrong one is the worst outcome. `NOT_OCCUPATION` is strictly distinct from the "99" unreviewed-abstain code. No feature presents an uncertain classification, a similarity edge, or a centrality score as stronger than it is. The graph is a *similarity* graph; betweenness identifies bridges between skill areas, not social popularity, and any surface showing it must say so.

Eval discipline. The 50-title dev set is contaminated — labels were revised after seeing predictions. Never quote it. The blind 100 is the real number.

Streamlit state. A write to a widget key must happen in a callback or above the widget. Any `st.rerun()` must have a stated terminating condition. Four bugs across P2.8c and P2.8d were variants of exactly this.

Live harnesses catch what unit tests cannot. Three chat bugs — deprecated `temperature=0` causing a hard 400, `max_tokens` starvation indistinguishable from a parse error, and a handle leaking into prose — were invisible to the unit suite and found only by `tools/check_chat_grounding.py`.

Canvas ordering. Read `highlight_index` from session state *above* `build_figure`, never from the panel's return value, or the previous answer's people light up beside the current answer's text.

Architecture. Two-layer module pattern throughout: a pure core with no Streamlit, no API key and full unit coverage, plus thin Streamlit glue. Established by `state.py`, `notes.py`, `summaries.py`, `assistant.py`. Architectural rules are enforced by AST checkers, not by grep and not by convention.

Environment. `python -m src.<module>` from repo root; `python -m pip`, never bare `pip`; `git --no-pager diff`; PowerShell `Select-String` and `Get-ChildItem -Recurse`, not `grep` and `find`.

8. Attribution

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